

Walmart Sales Analysis Report

Project Overview

This analysis was conducted to evaluate sales performance, customer purchasing behavior, payment trends, branch-level efficiency, customer satisfaction, and category profitability across Walmart branches.

The objective was to identify:

- Revenue-driving categories
- Underperforming branches
- Customer satisfaction risks
- Payment behavior trends
- Profitability opportunities

The analysis was performed using advanced PostgreSQL queries including:

- CTEs
 - Window Functions
 - Ranking Functions
 - CASE Statements
 - Aggregations
-

Executive Summary

The analysis revealed strong revenue concentration in a few categories and branches, while also identifying operational risks related to customer satisfaction and profitability.

Key Business Highlights

- Multiple payment methods were used, indicating diverse customer payment preferences.
 - Certain product categories generated strong sales but weak profit margins.
 - Some branches showed high revenue generation but below-average customer ratings.
 - Afternoon and evening shifts contributed the highest transaction volumes.
 - Revenue decline was observed in selected branches compared to the previous year.
-

Detailed Business Insights

1. Customer Payment Behavior Analysis

Objective

Identify the most preferred payment methods and purchasing behavior.

Analysis Performed

- Counted number of transactions by payment method
- Measured quantity sold across payment methods

Key Findings

- Digital payment methods accounted for the majority of transactions.
- Customers using e-wallets and credit cards contributed higher purchase quantities compared to cash users.
- Cash transactions were lower, indicating growing digital adoption.

Business Insight

Customers increasingly prefer fast and convenient digital payment systems.

Recommendation

- Increase cashback offers on digital payments.
 - Partner with payment wallets for promotions.
 - Improve checkout speed for digital transactions.
-

2. Highest Rated Product Categories by Branch

Objective

Identify customer-preferred product categories in each branch.

Analysis Performed

Used:

- AVG(Rating)
- Window Function (RANK)

Key Findings

- Each branch showed different category preferences.
- High-rated categories demonstrated stronger customer satisfaction and engagement.

Business Insight

Customer preferences vary significantly by location and branch demographics.

Recommendation

- Increase inventory for highly rated categories.
 - Use regionalized marketing strategies.
 - Replicate successful product mixes across branches.
-

3. Busiest Sales Day Analysis

Objective

Determine peak business days for each branch.

Analysis Performed

- Counted transactions by weekday
- Ranked busiest day branch-wise

Key Findings

- Weekend transaction volumes were significantly higher than weekdays.
- Some branches showed strong mid-week purchasing behavior.

Business Impact

Peak-day traffic creates opportunities for:

- upselling
- staffing optimization
- inventory planning

Recommendation

- Increase staffing during peak days.
 - Launch weekend promotional campaigns.
 - Optimize inventory replenishment before high-demand days.
-

4. Shift-Wise Sales Analysis

Objective

Identify transaction distribution across:

- Morning
- Afternoon
- Evening

Key Findings

Shift	Business Trend
Morning	Lowest transactions
Afternoon	High sales activity
Evening	Peak customer engagement

Business Insight

Afternoon and evening periods drive the majority of store traffic and revenue.

Recommendation

- Schedule experienced staff during evening hours.
 - Increase promotional displays during peak traffic.
 - Introduce evening combo offers.
-

5. Category Profitability Analysis

Objective

Identify categories generating:

- high revenue
- low profitability

Analysis Performed

Calculated:

- Total Revenue
- Profit Contribution
- Average Profit Margin

Key Findings

High-Risk Categories Identified

Some categories generated strong revenue but showed weak profitability.

Business Insight

High sales do not always translate into strong profits.

Potential causes:

- excessive discounting
- supply chain costs
- low-margin pricing

Recommendation

- Re-evaluate pricing strategies.
 - Optimize supplier contracts.
 - Promote high-margin product bundles.
 - Reduce dependency on low-margin categories.
-

6. Branch Performance vs Customer Satisfaction

Objective

Identify branches with:

- strong revenue
- poor customer experience

Analysis Performed

Combined:

- Total Revenue
- Average Customer Rating

Key Findings

Certain branches were categorized as:

High Sales – Low Satisfaction

Business Insight

These branches may face future:

- customer churn
- negative reviews
- declining loyalty

Even though short-term revenue remains strong.

Recommendation

- Investigate operational bottlenecks.
 - Improve customer service quality.
 - Reduce checkout waiting times.
 - Improve inventory availability.
-

7. Revenue Decline Analysis (2022 vs 2023)

Objective

Identify branches with the highest revenue decrease ratio.

Analysis Performed

Used:

- Year-over-Year revenue comparison
- Revenue decrease ratio calculation

Key Findings

Top 5 branches experienced measurable revenue decline compared to the previous year.

Business Insight

Revenue decline may indicate:

- increased competition
- reduced customer retention

- operational inefficiencies
- changing customer preferences

Recommendation

- Conduct branch-level root cause analysis.
 - Launch retention and promotional campaigns.
 - Review pricing and inventory strategies.
-

Stakeholder-Level Strategic Recommendations

Immediate Actions (0–3 Months)

- Improve customer experience in low-rated branches.
 - Increase digital payment incentives.
 - Optimize staffing during peak hours.
 - Focus on profitable product categories.
-

Medium-Term Actions (3–6 Months)

- Implement branch-specific sales strategies.
- Reduce low-margin dependency.
- Improve inventory forecasting.
- Launch customer loyalty initiatives.